

Simulating predictive-biomarker interactions in aggregated N-of-1 trials: a refined data-generating process following Hendrickson et al.

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Abstract

Background. The power of aggregated N-of-1 trials to validate a predictive biomarker is estimated by simulation. Hendrickson et al. (2020) introduced a framework for this, and were, to our knowledge, the first in the N-of-1 literature to encode the biomarker-by-treatment interaction as a correlation between the biomarker and the drug response within a joint multivariate normal distribution. We examine that encoding, correct its defects, and specify a refined data-generating process for trials of prazosin-like drugs.

Methods. We derive the moments and power of a paired-difference interaction statistic in closed form under the published process and refined versions of it. We verify them by Monte Carlo simulation, including runs of the published code. We compare covariance-based and mean-based binding of the biomarker, and stress-test analysis models against the features the refined process adds.

Results. As soon as any carryover is present, the published coupling rule gives every off-drug visit the full on-drug biomarker correlation. The interaction vanishes, and power falls to the test size at every positive half-life. Coupling in proportion to the remaining drug effect restores a smooth, provably monotone decline, and raised power in the published code from 0.13 to 0.64 at a half-life of 0.1 week. Mean-based binding that fades with the drug effect loses power almost identically. Under compound symmetry the joint distribution exists only for correlations below 0.256, so the published effect sizes of 0.3 and 0.6 relied on silently repaired matrices; separable AR(1) correlation raises the

bound to about 0.48. A biomarker that is also prognostic makes analyses that share the biomarker effect with baseline anticonservative.

Conclusions. The covariance-based encoding is coherent but needs graded coupling, a correlation structure with room for realistic effects, and validity checks that stop instead of repairing. We give the closed-form results that identify the power loss, the code changes for the published software, and a refined process for simulation studies of aggregated N-of-1 designs.

Keywords: N-of-1 trials; predictive biomarker; simulation; carryover; data-generating process; power; prazosin.

Background

Aggregated N-of-1 trials and predictive biomarkers

In an aggregated N-of-1 trial each participant moves between drug and placebo, or between drug and no drug, several times, and the individual series are analyzed together¹⁻³. Because every participant contributes both an on-drug and an off-drug response, the design can estimate how the drug effect varies between people, and so test whether a baseline measurement predicts who benefits^{4,5}. The motivating example is prazosin for post-traumatic stress disorder (PTSD). Higher pretreatment standing systolic blood pressure was associated with greater symptom reduction on prazosin but not on placebo⁶, which makes blood pressure a candidate predictive biomarker⁷⁻⁹.

Closed-form power calculations exist only for simplified versions of these designs, so power is estimated by simulating trials from a data-generating process (DGP) and analyzing each simulated trial¹⁰. The DGP decides what a simulation can show. In particular, it decides how the biomarker interaction is represented, how the drug effect builds and wanes, and how carryover from one period reaches the next.

The framework of Hendrickson et al.

Hendrickson et al.¹¹ (hereafter RH) introduced the simulation framework this paper builds on. Its contributions include:

- **A component decomposition of the symptom score.** The score is the baseline level minus three responses, each following a modified Gompertz curve:
 - a biological drug response (BR), driven by cumulative time on drug;
 - an expectancy response (PB), driven by time in the trial and scaled by the participant’s expectancy of receiving drug;
 - a time-varying natural course (TV).
- **Design machinery.** It covers open-label, crossover, blinded discontinuation and a Hybrid design combining all three. It tracks time on drug, time since discontinuation and expectancy for each visit.
- **Open software** (the `pmsimstats` R package).

RH also encoded the predictive biomarker in a distinctive way. Simulation studies of N-of-1 and crossover designs usually represent treatment-effect heterogeneity in one of two ways. The first is random participant-by-treatment variation^{4,12–14}. The second, for an observed moderator, is a biomarker-by-treatment term in the mean^{15–17}. RH instead placed the biomarker in the same multivariate normal distribution as the response components. They gave it a nonzero correlation c_{bm} with the drug response at on-drug visits and zero correlation elsewhere. A predictive biomarker is then one whose association with the outcome differs between on-drug and off-drug visits. To our knowledge this was the first covariance-based encoding of the biomarker-by-treatment interaction in the N-of-1 literature.

The encoding is natural for a biomarker that tracks individual variation in response rather than a fixed dose-response relation, and it lets a single correlation calibrate the strength of the effect against the variability of the drug response.

Why a follow-up is needed

Reusing RH’s process for a series of simulation studies exposed two problems. Both trace to the implementation, not to the idea.

1. **The joint distribution often does not exist.** Under the compound-symmetry structure RH used, the correlation matrix is positive definite only for c_{bm} below a ceiling, 0.256 in the Hybrid design without carryover. RH’s nonzero effect sizes were 0.3 and 0.6. The code repairs

any non-positive-definite matrix silently. At those effect sizes 37% of the published grid's matrices (20 of 54) were replaced by nearby valid matrices, whose correlations differ from the ones stated. At $c_{bm} = 0.6$ the repair lowered the on-drug correlation to 0.54 to 0.57.

2. Power collapses under any carryover. RH reported that power to detect the interaction fell sharply once carryover was introduced. In the Hybrid design at $N = 70$ and $c_{bm} = 0.3$ it fell from 0.74 to 0.12 at a half-life of 0.1 week, when only 0.1% of the drug effect remains a week after stopping. We show that this is an artifact of the coupling rule, not a property of the designs.

This paper therefore has three aims:

- to identify, in closed form, why power collapses under the published process;
- to give the changes to the published code that remove both problems;
- to specify a refined DGP, consistent with the pharmacology of short-half-life, titrated drugs such as prazosin, intuitive in its parameters and flexible in the features it can represent, for use across simulation studies of aggregated N-of-1 designs.

Methods

The published data-generating process

We use RH's notation, with a few additions. Participant i is observed at a baseline visit $t = 0$ and at post-baseline visits $t = 1, \dots, T$ (weeks 4, 8, 9, 10, 11, 12, 16 and 20 in the Hybrid design). The symptom score is

$$Y_{it} = \text{BL}_i - (TV_{it} + PB_{it} + BR_{it}),$$

so a larger response lowers the score. Each component has a mean given by the modified Gompertz curve

$$G(x; m, d, r) = m \frac{\exp(-de^{-rx}) - e^{-d}}{1 - e^{-d}},$$

evaluated at a component-specific clock:

- **BR** at cumulative time on drug t_{od} , with $m = 10.99$, $d = 5$, $r = 0.42$ and SD 8. The curve reaches its inflection at 3.8 weeks, half its maximum at 4.7 weeks and 90% at 9.2 weeks.
- **PB** at time in the trial, scaled by the expectancy e_t (1 in the open-label phase, 0.5 when blinded), with $m = 6.51$, $d = 5$, $r = 0.35$ and SD $10e_t$.
- **TV** at time in the trial, with the same curve as PB and SD 10.

Off drug, the drug-response mean decays from its previous value with half-life $t_{1/2}$: $\mu_{BR,t} = \mu_{BR,t-1} 2^{-t_{sd,t}/t_{1/2}}$, where t_{sd} is the time since discontinuation.

The vector (biomarker, baseline, $TV_{i.}$, $PB_{i.}$, $BR_{i.}$) is multivariate normal. Within each component the correlation between any two visits is $\rho = 0.8$ (compound symmetry). Between components it is 0.2 at the same visit and 0.1 otherwise. The biomarker's correlation with BR_{it} is $c_{bm} g_t$, where g_t is the *coupling*. RH's rule sets $g_t = 1$ wherever the drug-response mean is nonzero and 0 elsewhere. RH analyzed each trial with the linear mixed model

$$Y_{it} = \beta_0 + \beta_B B_i + \beta_D D_{it} + \beta_T t + \beta_{bm:D} B_i D_{it} + u_i + \varepsilon_{it},$$

with a random intercept u_i , the binary drug indicator D_{it} , and baseline included as a row with $D_{i0} = 0$. The interaction $\beta_{bm:D}$ is tested with Satterthwaite degrees of freedom.

Closed-form analysis

To isolate how carryover acts, we use a summary statistic whose distribution can be derived exactly (Appendix A). For each participant,

$$\Delta_i = \bar{Y}_i^{\text{on}} - \bar{Y}_i^{\text{off}} = \sum_t a_t Y_{it},$$

the difference between the mean on-drug and mean off-drug scores, which is regressed on B_i by least squares. We call it E9 when it uses post-baseline visits only, and E9b when baseline is counted as one more off-drug visit, as in RH's analysis. Monte Carlo simulation of RH's mixed model shows that E9b reproduces its mean estimates to within 0.003 and its power closely (Results). The derivation gives three results.

- **Proposition 1.** The variance of Δ does not depend on the carryover half-life. The half-life enters only through the response means and the coupling, and neither appears in the variance.

- **Proposition 2.** The population slope is

$$\beta_{bm:D}^{\text{E9}} = -c_{bm} \frac{\sigma_{BR}}{\sigma_{bm}} (1 - \bar{g}_{\text{off}}), \quad \beta_{bm:D}^{\text{E9b}} = -c_{bm} \frac{\sigma_{BR}}{\sigma_{bm}} \left(1 - \frac{n_{\text{off}}}{n_{\text{off}} + 1} \bar{g}_{\text{off}}\right),$$

where $\bar{g}_{\text{off}}$ is the mean coupling over the n_{off} post-baseline off-drug visits.

- **Power** is a noncentral- t probability governed by the correlation between Δ and B . That correlation is proportional to the *coupling gap* $1 - \bar{g}_{\text{off}}$.

So carryover can affect power only through the coupling gap. The propositions reduce the question of why power collapses to the question of what the coupling rule does to $\bar{g}_{\text{off}}$.

The ceiling. Write the response block of the correlation matrix as M and the biomarker's correlations with the response components as $c_{bm}u$. The full matrix is positive definite if and only if M is, and

$$c_{bm}^2 u^\top M^{-1} u < 1.$$

The left side is the squared multiple correlation of the biomarker on the whole response trajectory, so the ceiling on c_{bm} is $(u^\top M^{-1} u)^{-1/2}$ (Appendix A). It depends on the correlation structure, the design and the coupling profile.

Refinements

Each refinement removes one defect (Table 1). The first three are corrections. The rest are modeling choices, and each can be switched off to recover the published process.

Table 1: Refinements to the published process.

Step	Change	Defect removed or feature added
1	Stop when the correlation matrix is not positive definite	Silent repair changes the correlations, and so the effect, without warning

Step	Change	Defect removed or feature added
2	Graded coupling: $g_t = 1$ on drug, $2^{-t_{sd,t}/t_{1/2}}$ off drug after exposure, 0 before any exposure	Step coupling removes the interaction at any positive half-life
3	Anchored carryover: decay the drug-response mean from its last on-drug value	The recursion compounds the decay at the second and later off-drug visits
4	AR(1) correlation within each component, $\rho^{ t-s }$ in weeks	Compound symmetry overstates long-lag correlation
5	Separable cross-component structure, $c_1 \rho^{ t-s }$	Raises the ceiling; gives a valid factored generator
6	Options: prognostic biomarker couplings, phase-specific measurement error, decay form, dropout	Not representable in the published process

Pharmacological grounding

The refined process is specified for short-half-life, titrated drugs, with prazosin as the reference case.

- **Time scales.** Prazosin's plasma half-life is a few hours, against weekly visits. The effect of a dose therefore wanes within days, and pharmacological carryover at the visit scale is short. We take half-lives of 0.1 to 0.2 week as the pharmacological range and regard longer carryover as behavioral or psychological, to be simulated as a labeled sensitivity scenario.
- **Titration and onset.** The drug response follows the Gompertz curve in time on drug, which reflects dose titration and the accumulation of benefit. Its landmarks (half of maximum at 4.7 weeks) should be read clinically when the parameters are chosen.
- **Re-exposure.** RH's process restarts time on drug at zero when the drug is resumed. For prazosin this is justified, because the drug is re-titrated after an interruption. For drugs without titration, the response should instead rebuild from its residual level. This is an option, not a default.
- **Coupling.** Graded coupling makes the biomarker's association with the drug response fade at the rate the drug effect fades. That is the pharmacological reading of a predictive biomarker: it predicts the size of the drug effect, so it cannot predict a drug effect that is no longer present.

Covariance-based and mean-based binding

The biomarker can be bound to the drug effect in the covariance, as RH did, or in the mean. Under mean moderation, the on-drug response is shifted by $\beta_{bm} b_i \sigma_{BR} h_t$, where b_i is the standardized biomarker and h_t the exposure. A third form, proposed for this compendium, generates the drug response from a participant-level sensitivity,

$$BR_{it} = \omega_i f(t) + \varepsilon_{it}, \quad \omega_i = \bar{\omega} (1 + \zeta b_i) + u_i,$$

where $f(t)$ is the exposure-response time course, ζ the biomarker's effect on sensitivity, and u_i the sensitivity it does not explain. Its covariance follows from the model, so no ceiling arises, and the off-drug variance of the drug response shrinks with the response. We compare the three forms in the Discussion.

Simulation studies

Three sets of simulations verify the results. All are reproducible from the scripts listed in the Availability section.

- 1. Closed-form verification.** 5,000 Monte Carlo replicates per cell of the paired-difference statistic, and 500 to 1,000 replicates of RH's mixed model, in the Hybrid, crossover and open-label plus blinded discontinuation designs at $N = 70$.
- 2. The published code itself.** RH's `generateData()` at commit 58b32a9 was run twice on its own pipeline: as published, and with only the coupling lines replaced by graded coupling (Appendix B). We used the Hybrid design, $N = 70$, $c_{bm} = 0.25$, no censoring, 500 replicates per cell, and the same seeds for both versions.
- 3. A pre-registered stress test of analysis models** against the refined process. It has 39 cells and 26,500 simulated trials. Correlation structure, sample size, carryover, decay form, dropout, measurement error and a prognostic biomarker were each varied from a base cell (separable AR(1) at $\rho = 0.7$, $N = 70$, $t_{1/2} = 0.5$ week). Size was judged at 1,000 null replicates per cell.

Table 2: Closed-form power by carryover half-life (weeks). Hybrid design, $N = 70$, $c_{bm} = \beta_{bm} = 0.25$, compound symmetry. Monte Carlo estimates agree within simulation error.

Coupling rule	Statistic	0	0.1	0.2	0.5	1	2
Step (published)	E9	0.721	0.050	0.050	0.050	0.050	0.050
	E9b	0.654	0.081	0.081	0.081	0.081	0.081
Graded	E9	0.721	0.721	0.712	0.631	0.473	0.278
	E9b	0.654	0.653	0.647	0.586	0.473	0.328
Mean moderation	E9	0.681	0.681	0.682	0.682	0.682	0.682
	E9b	0.619	0.619	0.620	0.620	0.621	0.621

Results

The published coupling rule removes the interaction

Under RH's rule every off-drug visit in the Hybrid design follows an exposure. With any positive half-life its drug-response mean is positive, so its coupling is 1. Hence $\bar{g}_{\text{off}} = 1$ and, by Proposition 2,

$$\beta_{bm:D}^{\text{E9}} = 0, \quad \beta_{bm:D}^{\text{E9b}} = -c_{bm} \frac{\sigma_{BR}}{\sigma_{bm}} \frac{1}{n_{\text{off}} + 1} \quad \text{for every } t_{1/2} > 0.$$

Power is a step function of the half-life (Table 2, Figure 1). It takes its no-carryover value at $t_{1/2} = 0$ and a constant at any positive half-life, whether 0.01 week or 10. For E9b that constant is 0.081, a residue of the baseline row, which alone keeps zero coupling. The published analysis gives the same: the mean estimate falls from -0.126 to between -0.026 and -0.032 at every positive half-life, and power from 0.63 to 0.08 to 0.11.

With constant coupling, the biomarker moderates the on-drug and off-drug responses equally, so there is no interaction left to detect. RH's finding that small carryover destroys power is therefore a property of the coupling rule, not of carryover or of the designs.

Graded coupling restores a smooth decline

Under graded coupling the mean off-drug coupling is

$$\bar{g}_{\text{off}}(t_{1/2}) = \frac{1}{n_{\text{off}}} \sum_{t_{\text{off}}} 2^{-t_{sd,t}/t_{1/2}},$$

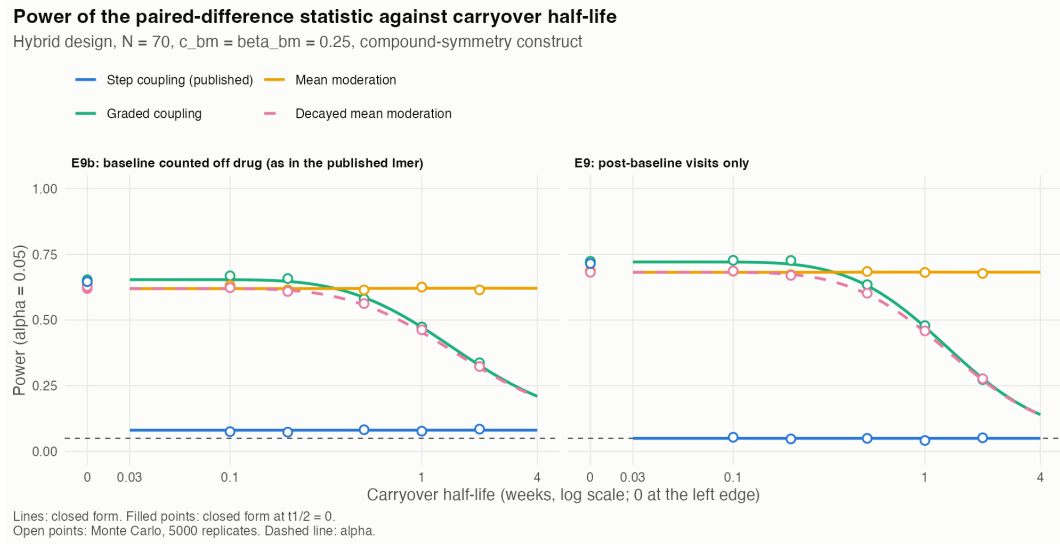


Figure 1: Closed-form power (lines) against carryover half-life, with Monte Carlo estimates (points), under the published step rule, graded coupling, mean moderation and decayed mean moderation. Hybrid design, $N = 70$, compound symmetry.

which is strictly increasing in $t_{1/2}$, because every off-drug visit has $t_{sd} \geq 1$ week. Two consequences follow.

- **Power is strictly decreasing in the half-life** (Proposition 3, Appendix A).
- **The loss vanishes faster than any power of the half-life as $t_{1/2} \rightarrow 0$.** At 0.1 week, $\bar{g}_{\text{off}} \leq 2^{-10}$ and power is unchanged to within 0.001.

The published rule is the limit of graded coupling as $t_{1/2} \rightarrow \infty$, applied at every positive half-life.

Power depends on the coupling profile only through the coupling gap $1 - \bar{g}_{\text{off}}$ (Figure 2). The gap is 1 without carryover; under graded coupling it is 0.991 at 0.2 week, 0.906 at 0.5 and 0.754 at 1. Under the published rule it is 0 at every positive half-life. Because the noncentrality is proportional to the gap, the sample size needed for a given power scales roughly as $1/\text{gap}^2$.

The correction in the published code

Replacing only RH's coupling lines with graded coupling (Appendix B) and running their pipeline unchanged gave the following results (Table 3). The two versions coincide without carryover, as they must. At the published half-lives the graded version keeps the full interaction, and at 1 week it shows a gradual decline. No matrix was repaired in either version, because $c_{bm} = 0.25$ is below the ceiling.

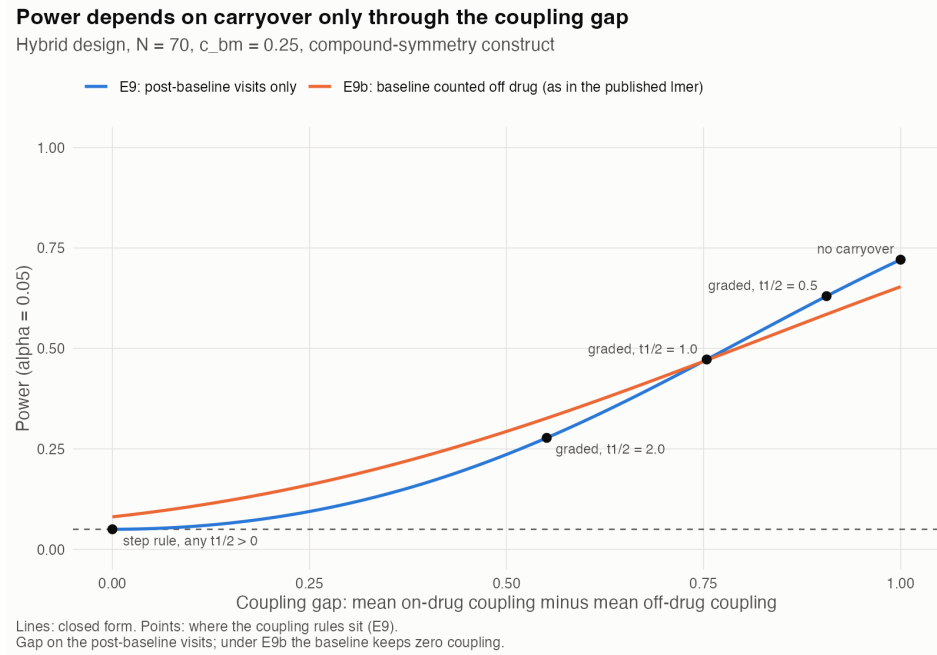


Figure 2: Closed-form power against the coupling gap, with the positions of the coupling rules. Hybrid design, $N = 70$, $c_{bm} = 0.25$.

Table 3: RH's code at commit 58b32a9, as published and with graded coupling: Hybrid design, $N = 70$, $c_{bm} = 0.25$, 500 replicates per cell, same seeds. Monte Carlo SE of power at most 0.022.

$t_{1/2}$ (weeks)	Mean estimate, published	Mean estimate, graded	Power, published	Power, graded
0	-0.131	-0.131	0.668	0.668
0.1	-0.028	-0.129	0.126	0.638
0.2	-0.029	-0.131	0.092	0.656
1	-0.033	-0.107	0.094	0.462

Table 4: Largest feasible biomarker correlation c_{bm} under graded coupling, without carryover, minimum over the Hybrid, crossover and open-label plus blinded discontinuation designs. Under the separable form at $\rho = 0.7$ the Hybrid value is 0.480.

ρ	Compound symmetry	AR(1), decaying cross-correlation	Separable AR(1)
0.8	0.256	0.291	0.440
0.7	0.346	0.452	0.474

Room for realistic effect sizes

The ceiling $(u^\top M^{-1}u)^{-1/2}$ depends strongly on the correlation structure (Table 4).

- **Compound symmetry.** The correlation is the same at every lag, so the visits share a large person-level component that the biomarker can correlate with only once. Under that structure the ceiling is 0.256 in the Hybrid design.
- **AR(1).** Less is shared between distant visits, so a single correlation repeated across many visits demands more of the biomarker. AR(1) correlation alone therefore lowers the ceiling.
- **Separable form.** The cross-component correlation follows the same AR(1) decay ($c_1\rho^{|t-s|}$). This makes the response block a Kronecker product, and the ceiling rises to about 0.48 at $\rho = 0.7$.

A biomarker correlation of 0.45 is feasible in every design under the separable form, and not under the published structure. Effects as large as RH’s 0.6 exceed every covariance-based ceiling in the Hybrid design. They can be represented only by binding the biomarker in the mean.

Consequences for the analysis

The refined process can simulate features the published one cannot, and these change which analysis is safe. Two results from the stress test bear on the DGP directly.

A prognostic biomarker. Suppose blood pressure is associated with the natural course or with placebo response as well as with the drug effect. We simulated a correlation of 0.3 with the natural-course component (constant over visits, or growing along its curve) or with the placebo component. Analyses that keep baseline as an outcome row use one biomarker main effect β_B for baseline and for the off-drug visits. When the biomarker is prognostic, those two associations differ, and part of the difference is attributed to the interaction (Appendix A.4).

The published random-intercept model rejected a true null 0.107 to 0.159 of the time. An unstructured-covariance mixed model rejected up to 0.42. Analyses that free the biomarker's baseline association held the nominal level (0.040 to 0.063). One of them adds expectancy and a biomarker-by-expectancy term to the mean model; another adds the term `bmc:b10`, which gives baseline its own biomarker slope.

Size more generally. Across 11 non-prognostic null cells:

- the published analysis rejected 0.069 on average, and up to 0.081 at $N = 35$;
- an unstructured model with Kenward-Roger inference¹⁸ reached 0.087 at $N = 35$;
- random-intercept models with continuous-time AR(1) residuals and CR2 cluster-robust standard errors^{19,20} stayed between 0.036 and 0.069.

A simulation that omits a prognostic pathway cannot reveal these failures. The refined process includes it as an option for that reason.

Discussion

Covariance-based and mean-based binding

RH's covariance-based encoding has real strengths.

- **One interpretable parameter.** The interaction strength is a single correlation, interpretable as the share of drug-response variation the biomarker explains.
- **A reduced form of responder heterogeneity.** If participants fall into latent responder classes and the biomarker is associated with class membership, the second moments of the resulting mixture are those of the covariance encoding, with c_{bm}^2 equal to the product of the between-class variance fractions of the biomarker and the drug response²¹. The encoding therefore approximates a responder-class mechanism without requiring the mixture to be identified. Identifying it needs samples far larger than an N-of-1 program has. The approximation has limits: no two-class mixture reaches the covariance encoding exactly, because a mixture always moves the conditional mean as well.
- **The biomarker explains variance rather than adding it.** At the same slope this gives slightly more power than mean moderation without carryover (0.721 against 0.681 for E9).

- **It fits a biomarker that marks individual sensitivity to the drug** better than one that sets a fixed dose-response shift.

Its weaknesses, all made explicit by this analysis, are structural.

- **The joint distribution must exist**, which bounds the effect size below values that are clinically plausible. The bound depends on the correlation structure and the design.
- **The coupling must be specified at off-drug visits**, and the natural choice of an indicator is wrong under carryover.
- **The drug-response component keeps its full variance at off-drug visits**, with zero mean. This adds noise to every on-off contrast, and it is the least defensible feature of the published construct.
- **A correlation is less intuitive than a mean effect for clinical readers**. The slope it implies, $-c_{bm}\sigma_{BR}/\sigma_{bm}$, must be derived before it can be compared with an observed effect such as 14 points of symptom change per 10 mm Hg⁶.

Mean moderation has no ceiling, an intuitive parameter (points of symptom change per SD of biomarker) and no coupling to specify off drug. In its plain form the shift is applied only at on-drug visits, so its power is flat in the half-life (0.681 at every value for E9). But it adds variance instead of explaining it, and that flatness is a coding choice, not a finding: the biomarker's effect stops the moment the drug is stopped, while the drug effect itself carries over.

When the mean shift is decayed with the drug effect, as graded coupling decays the correlation, power declines almost exactly as it does under graded coupling: 0.681, 0.681, 0.674, 0.598, 0.454 and 0.271 at half-lives of 0, 0.1, 0.2, 0.5, 1 and 2 weeks, against 0.721, 0.721, 0.712, 0.631, 0.473 and 0.278. How much carryover costs is therefore decided by whether the biomarker's effect fades with the drug effect, not by whether it is bound in the covariance or in the mean. Comparisons that report larger carryover losses under covariance binding than under mean binding are comparing a fading interaction with a non-fading one.

The sensitivity model combines the strengths. It binds the biomarker in the mean through individual sensitivity, so it has no ceiling. Its moderation is proportional to the response and fades with it. Its off-drug variance shrinks with the response. Its implied covariance reproduces the covariance-based encoding's qualitative behavior. It is not yet implemented, and its response half-life for prazosin in PTSD is not known.

Recommendations

For simulation studies that build on RH, we recommend the following.

1. **Always apply the corrections:** stop on non-positive-definite matrices, use graded coupling, and use anchored carryover.
2. **Use the separable AR(1) structure** at a stated ρ , and choose c_{bm} below the ceiling for every design in the study.
3. **Keep RH's process as a reference arm** whenever results are compared with the published ones. Attribute each difference to a named refinement.
4. **Report effect sizes on two scales:** as c_{bm} , and as the implied slope in outcome points per SD of biomarker.
5. **Include a prognostic scenario** whenever the analysis shares the biomarker effect with the baseline visit.
6. **Use mean moderation, or the sensitivity model once available, for effects above the covariance ceiling,** and as a sensitivity analysis on the encoding itself.

Limitations

- **The closed form covers one statistic.** It is exact for the paired-difference statistic, not for RH's mixed model. The two agree closely in the Hybrid design, but in the crossover design the closed form understates what a time-adjusted analysis achieves.
- **Parameter values are not estimated.** The Gompertz parameters, correlations and prognostic couplings are RH's values or chosen values, not estimates from prazosin trial data.
- **The process is Gaussian with a single biomarker.**
- **The pharmacological carryover range** for prazosin at weekly visits is inferred from its plasma half-life, not measured as a symptom-response half-life.

Conclusions

RH introduced a covariance-based encoding of the predictive-biomarker interaction that remains a coherent choice for simulating aggregated N-of-1 trials. Its published implementation removes the interaction whenever there is carryover, and it silently alters correlations it cannot represent. Both

problems have exact explanations and small fixes. With graded coupling, a separable correlation structure and validity checks that stop instead of repairing, the process behaves as the pharmacology of a short-half-life drug suggests it should: power declines smoothly, and negligibly at pharmacological half-lives. The refined process, with options for prognostic pathways and other features that change which analysis is valid, is a sound reference for simulation studies of these designs.

Abbreviations

BR: biological drug response; CO: crossover; CR2: bias-reduced cluster-robust variance estimator; CS: compound symmetry; DGP: data-generating process; OL+BDC: open label followed by blinded discontinuation; PB: expectancy (placebo-belief) response; PTSD: post-traumatic stress disorder; RH: Hendrickson et al. (2020); TV: time-varying natural course.

Declarations

Ethics approval and consent to participate. Not applicable: the study uses simulated data only.

Consent for publication. Not applicable.

Availability of data and materials. All code and simulated data are in the pmsimstats-ng repository:

- the closed-form calculations, in `analysis/scripts/quick-sim/carryover-closed-form/01-e9-closed-form`
- the check of the published code, in `analysis/scripts/quick-sim/hendrickson-problems/14-graded-patc`
- the stress test, in `analysis/scripts/quick-sim/carryover-closed-form/11-strawman-stress-test.R` and `12-strawman-stress-decision.R`;
- the published code, at `github.com/rchendrickson/pmsimstats`, commit 58b32a9.

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Acknowledgements. To be completed by the authors.

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Appendix A. Closed-form derivations

A.1 Moments of the paired-difference statistic

Let $a_t = 1/n_{\text{on}}$ at on-drug visits and $-1/n_{\text{off}}$ at off-drug visits, so that $\Delta_i = \sum_t a_t Y_{it}$. Because the weights sum to zero, the baseline level cancels and $\Delta = -\sum_t a_t S_t$, with $S_t = TV_t + PB_t + BR_t$. With $m_1 = \sum a_t$, $m_2 = \sum a_t^2$, $m_e = \sum a_t e_t$, $m_{ee} = \sum a_t^2 e_t^2$ and $m_{2e} = \sum a_t^2 e_t$, the variance under compound symmetry is

$$\begin{aligned} \text{Var}(\Delta) = & (\sigma_{TV}^2 + \sigma_{BR}^2)[\rho m_1^2 + (1 - \rho)m_2] + \sigma_{PB}^2[\rho m_e^2 + (1 - \rho)m_{ee}] \\ & + 2\sigma_{TV}\sigma_{BR}[c_{\times}m_1^2 + (c_1 - c_{\times})m_2] + 2(\sigma_{TV} + \sigma_{BR})\sigma_{PB}[c_{\times}m_1m_e + (c_1 - c_{\times})m_{2e}], \end{aligned}$$

where c_1 and $c_{\times}$ are the same-visit and other-visit cross-component correlations.

- **Proposition 1.** No term depends on $t_{1/2}$, so $\text{Var}(\Delta)$ is free of the half-life. The argument applies to any construct in which the half-life does not enter the response block.
- **Proposition 2.** Only BR_t is correlated with B , with covariance $c_{bm}g_t\sigma_{bm}\sigma_{BR}$, so $\beta_{bm:D} = -c_{bm}(\sigma_{BR}/\sigma_{bm})\sum_t a_t g_t$. With $g_t = 1$ on drug, this gives the expressions in the Methods.
- **Residual variance.** Under covariance binding, $\tau^2 = \text{Var}(\Delta) - \beta_{bm:D}^2\sigma_{bm}^2$. Under mean binding the shift adds variance and $\tau^2 = \text{Var}(\Delta)$.

A.2 Power

Within a path of n participants, the least-squares slope is exactly normal given the biomarker values. Its t -statistic is exactly noncentral t with $n - 2$ degrees of freedom and noncentrality $\beta_{bm:D}\sqrt{\text{SS}_B}/\tau$, where $\text{SS}_B \sim \sigma_{bm}^2\chi_{n-1}^2$.

At $\text{SS}_B = \sigma_{bm}^2(n - 1)$ the noncentrality is $\Lambda = \sqrt{n - 1}\rho_{\Delta B}/\sqrt{1 - \rho_{\Delta B}^2}$, where $\rho_{\Delta B} = \beta_{bm:D}\sigma_{bm}/\text{SD}(\Delta)$ is proportional to the coupling gap.

Pooling paths with a common intercept adds the between-path variation in mean contrast to the residual variance; in the Hybrid design it is under 0.7% of the total. Power is the noncentral- t tail probability averaged over the χ^2 law of SS_B .

A.3 Monotonicity under graded coupling

$$\frac{d\bar{g}_{\text{off}}}{dt_{1/2}} = \frac{\ln 2}{n_{\text{off}} t_{1/2}^2} \sum_{t_{\text{off}}} t_{sd,t} 2^{-t_{sd,t}/t_{1/2}} > 0.$$

- **Proposition 3.** By Proposition 2 the slope's magnitude is strictly decreasing in $t_{1/2}$. By A.1, τ^2 is strictly increasing, since $\text{Var}(\Delta)$ is fixed. So the noncentrality and the power are strictly decreasing.
- **Small half-lives.** $\bar{g}_{\text{off}} \leq 2^{-1/t_{1/2}}$, which gives the behavior near zero stated in the Results.
- **Large half-lives.** $\bar{g}_{\text{off}} \rightarrow 1$ as $t_{1/2} \rightarrow \infty$, which recovers the step rule.

A.4 The ceiling and the shared biomarker effect

The ceiling. Order the variables as biomarker, then responses, and partition the correlation matrix as $\begin{pmatrix} 1 & c_{bm}u^\top \\ c_{bm}u & M \end{pmatrix}$. By the Schur complement, the matrix is positive definite if and only if M is positive definite and $1 - c_{bm}^2 u^\top M^{-1} u > 0$. Hence the ceiling is $c_{bm}^* = (u^\top M^{-1} u)^{-1/2}$.

The shared biomarker effect. Let s_0 , s_{off} and s_{on} be the true slopes of Y on B at baseline, at post-baseline off-drug visits and at on-drug visits. A model with baseline as an off-drug row estimates one β_B for s_0 and s_{off} .

Under a prognostic biomarker with no interaction, $s_0 = 0 \neq s_{\text{off}} = s_{\text{on}} = s_{\text{prog}}$. The fitted $\hat{\beta}_B \approx f s_{\text{prog}}$ for some $0 < f < 1$, set by the weight the baseline row receives. The interaction absorbs the remainder: $\hat{\beta}_{bm:D} \approx (1 - f) s_{\text{prog}} \neq 0$.

Adding $B_i \cdot \mathbf{1}\{t = 0\}$ frees s_0 and removes the bias. The expectancy term $B_i \cdot e_t$, with $e_0 = 0$, has the same effect.

Appendix B. Code changes

B.1 Graded coupling in RH's `generateData()` (commit 58b32a9)

The published lines (`R/generateData.R`, lines 131 to 138) couple the biomarker wherever the drug-response mean is nonzero:

```

# correlation with biomarker
for(p in 1:nP){
  n1<-paste(trialdesign$timeptname[p], "br", sep=".")
  if(means[which(n1==labels)]!=0){
    correlations[n1, 'bm']<-modelparam$c.bm
    correlations['bm', n1]<-modelparam$c.bm
  }
}

```

420 The replacement couples in proportion to the remaining drug effect, with no coupling before first
 421 exposure:

```

# correlation with biomarker: full on drug, decaying with the drug
# effect off drug (graded coupling), zero before first exposure
for(p in 1:nP){
  n1<-paste(trialdesign$timeptname[p], "br", sep=".")
  if(d[p]$tod>0){
    g<-1
  }else if(d[p]$tsd>0 && modelparam$carryover_t1half>0){
    g<-(1/2)^(d[p]$tsd/modelparam$carryover_t1half)
  }else{
    g<-0
  }
  correlations[n1, 'bm']<-modelparam$c.bm*g
  correlations['bm', n1]<-modelparam$c.bm*g
}

```

422 B.2 Stop instead of repairing

423 The published code replaces a non-positive-definite matrix with a nearby valid one (lines 145 to
 424 149). The replacement stops, so that an infeasible cell is reported rather than silently changed:

```

# published:
# if(!is.positive.definite(sigma)){
#   sigma<-make.positive.definite(sigma, tol=1e-3)
# }
if(!corpcor::is.positive.definite(sigma)){
  stop(sprintf('c.bm = %.3f exceeds the feasible range for this design',
              modelparam$c.bm))
}

```

B.3 Anchored carryover on the drug-response mean

The published recursion (lines 85 to 89) decays the previous visit's mean by the total time since discontinuation. At the second and later off-drug visits that mean has already been decayed, so the decay compounds. The replacement decays from the last on-drug value:

```

brmeans<-modgompertz(d$tod,rp$max,rp$disp,rp$rate)
if(nP>1){
  last_on<-0
  for(p in 1:nP){
    if(d[p]$onDrug){
      last_on<-brmeans[p]
    }else if(d[p]$tsd>0 && modelparam$carryover_t1half>0){
      brmeans[p]<-last_on*(1/2)^(d$tsd[p]/modelparam$carryover_t1half)
    }
  }
}

```

B.4 The exposure-decayed drug indicator in the analysis

Analyses that use an exposure-decayed drug indicator must code off-drug visits before first exposure as unexposed. The time since discontinuation is zero there, and the decay function returns 1 at zero, so without a guard a placebo-first crossover participant would be coded fully exposed at every

433 visit. In the pmsimstats-ng carryover manuscript this guard was missing. Restoring it raised the
434 power of the exposure-weighted analysis in the crossover design from 0.51 to 0.81:

```
exposure_weight <- function(Db, tsd, halflife) {  
  w <- 0.5^(tsd / halflife)  
  ifelse(Db == 1, 1, ifelse(tsd > 0, w, 0))  
}
```

435 B.5 The refined process in pmsimstats-ng

436 The package function buildSigma() (R/generateData.R) implements graded coupling, with `c.bm`
437 on drug, `c.bm * exp(-lambda_cor * tsd)` off drug after exposure and zero before exposure. It
438 also implements the AR(1) structures and the placebo coupling `c.bm.pb`. The separable form,
439 anchored carryover, the stop on infeasible matrices, and the natural-course and measurement-error
440 options are implemented in the simulation drivers of the stress test and are to be moved into the
441 package.